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MILITARY STANDARD

ORGANIC SOLVENTS, SPECTROPHOTOMETRIC GRADE

FSC 6810

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DEPARTMENT OF DEFENSE
Washington, DC 20301

Organic Solvents, Spectrophotometric Grade

MIL-STD-1219

1. This Military Standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Munitions and Chemical Command, Attn: DRSMC-TSC-S (A), Aberdeen Proving Ground, MD 21010, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FOREWORD

This is the first book format military standard generated on organic solvents, spectrophotometric grade.

This standard is approved for use by all departments and agencies of the Department of Defense in the selection of items for application. It is intended to prevent the entry of unnecessary items (sizes, types, varieties) into the Department of Defense logistics system. This document is not intended to restrict any service in selecting new items resulting from state-of-the-art changes.

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1. SCOPE

1.1 Coverage. This standard is a presentation of organic solvents, spectrophotometric grade, including ACS grade. This standard does not necessarily include all of the items represented by the title or all those items which are commercially available. It does contain items preferred for use in the selection of organic solvents, spectrophotometric grade, for application by the Department of Defense.

1.2 Application. Organic solvents, spectrophotometric grade, are used in ultraviolet (UV) and infrared (IR) spectrophotometric methods for the identification and analysis of dissolved substances.

2. REFERENCED DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this standard to the extent specified herein.

SPECIFICATIONS

FEDERAL

O-C-265	- Chemicals, Analytical, General Specification For
PPP-C-2020	- Chemicals, Liquid, Dry And Paste; Packaging Of

STANDARDS

FEDERAL

FED-STD-313	- Material Safety Data Sheets, Preparation And The Submission Of

MILITARY

MIL-STD-12	- Abbreviations For Use On Drawings, And In Specifications, Standards And Technical Documents

PUBLICATIONS

DOD 4145.19-R-1	- Storage And Materials Handling
DOD 4160.21-M	- Defense Utilization And Defense Disposal Manual
DOD 6050.5-LR	- DOD Hazardous Materials Information System, Hazardous Item Listing

TECHNICAL BULLETINS

TB MED 502 (DLAM 1000.2)	- Occupational And Environmental Health Respiratory Protection Program
TB MED 506	- Occupational And Environmental Health Occupational Vision

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should

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be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this standard to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN CHEMICAL SOCIETY (ACS)

Reagent Chemicals, American Chemical Society Specifications

(Application for copies should be addressed to the American Chemical Society, 1155 Sixteenth Street, NW, Washington, DC 20036.)

ASTM

ASTM E 380 - Standard For Metric Practice

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

US PHARMACOPEIAL CONVENTION, INCORPORATED

The United States Pharmacopeia
National Formulary

(Application for copies should be addressed to Mack Publishing Company, Easton, PA 18042.)

CODE OF FEDERAL REGULATIONS (CFR)

Title 29 - Department of Labor, Occupational Safety and Health Agency; General Industry Standards and Interpretations
Title 40 - Environmental Protection Agency; Hazardous Waste And Consolidated Permit Regulations
Title 49 - Department Of Transportation; Hazardous Materials Regulations

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

National Fire Codes

(Application for copies should be addressed to National Fire Protection Association, Battery March Park, Quincy, MA 02269.)

Technical society and technical association specifications, standards, and publications are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. GLOSSARY

3.1 Definitions.

Absorbance (A) - The logarithm to the base 10 of the reciprocal of transmittance (T), expressed as: $A=\log 1/T$.

Absorption spectra - A graphic representation of absorbance or any function of absorbance plotted against wavelength or a function of wavelength.

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Absorptivity (a) - The quotient of the absorbance (A) divided by the product of the concentration (c) of the substance in grams per liter, and the absorption path length (b) in centimeters. It is expressed as: $a=A/bc$. In practice, it is the absorbance of a solution containing 1 gram per liter of solute in a cell with an absorption path of 1 centimeter.

Molar absorptivity (ε) - The quotient of the absorbance (A) divided by the product of the concentration of the solute expressed in moles per liter, and the absorption path length in centimeter. It is also the product of the absorptivity (a) and the molecular weight of the substance (MW). It is expressed as: $\varepsilon=a(MW)=A(MW)/bc$.

Radiant power (P, P_0) - The energy of radiation reaching a given area of a detector per second.

Transmittance (T) - The quotient of the radiant power transmitted by a sample divided by the radiant power incident upon a sample, expressed as: $T=P/P_0$. In practice, it is the ratio of the radiant power transmitted through a sample to the radiant power transmitted through a reference material. It is calculated as percent transmittance.

Wavelength (WL) - The linear distance traversed by one complete wave cycle.

Wave number - The number of wave cycles in one centimeter for an electromagnetic wave.

3.2 Abbreviations. The use of abbreviations shall be in accordance with MIL-STD-12. Metric system abbreviations and symbols shall be in accordance with ASTM E 380.

4. GENERAL REQUIREMENTS

4.1 Packaging data and labeling. All chemicals included in this standard shall be packaged in accordance with Federal Specification PPP-C-2020 and all applicable documents referenced therein. Shipping containers shall be labeled in accordance with current Department of Transportation (DOT) Hazardous Materials Regulations applicable to each chemical.

4.2 Hazardous materials information. DOD 6050.5, DOD Hazardous Materials Information System (HMIS) requires Material Safety Data Sheets (MSDSs) for all hazardous materials used by DOD. The contractual acquisition of a MSDS is accomplished through use of Federal Acquisition Regulation, paragraph 52.223-3, Hazardous Material Identification and Material Safety Data. The MSDS is prepared in accordance with the instructions in FED-STD-313.

4.3 Safety.

4.3.1 Personal protective measures. The necessary respiratory, eye and skin protection to be used when handling chemicals shall be prescribed by the responsible installation industrial hygiene, medical and safety authorities.

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4.3.1.1 Respiratory protection. Respirators, approved by the National Institute for Occupational Safety and Health (NIOSH) or the Mine Safety and Health Administration (MSHA) or by particular respirator schedules of the Bureau of Mines (BM) for the compounds being used, may be employed for intermittent exposure or for supplementing other control measures (refer to TB MED 502 or DLAM 1000.2). Ventilation should be adequate to remove hazardous concentrations.

4.3.1.2 Skin protection. Personnel using these compounds shall be provided with and required to use impervious gloves, sleeves, aprons, and boots whenever indicated. Protective creams and ointments commonly known as "barrier creams" may be of value in certain cases. However, barrier creams should not be used to replace protective clothing. Eye lavages and emergency showers should be located where there is a potential for direct contact with harmful chemicals.

4.3.1.3 Face and eye protection. Personnel using these compounds shall be provided with and required to wear chemical splash-proof safety goggles. (Refer to TB MED 506.) In addition, face shields shall be provided and worn over the goggles if splashing could occur.

4.3.1.4 Training. Training should be provided to ensure that employees know potential hazards of the chemicals with which they come in contact and the symptoms of over-exposure as well as how these chemicals affect the body and bodily functions. A person or persons shall be adequately trained to render first aid.

4.3.1.5 Exercises. Participation in exercises should be stressed to demonstrate skills in the use of personal protective equipment and emergency response equipment.

4.3.2 Storage conditions. DOD 4145.19-R-1 describes general storage practices and requirements for hazardous materials in the DOD supply system. Specific requirements provided in the following paragraphs are supplementary in nature and should be observed in consonance with the DOD storage regulations.

4.3.2.1 Flammable, combustible, pyrophoric and ignitable materials. A flammable material is generally any solid, liquid, vapor or gas that ignites easily and burns rapidly. Combustible materials are generally those that are difficult to ignite and burn slowly. The DOT, in Section 173.115, Subpart D, Part 173 of 49CFR, 1982 Revised and as amended, defines a flammable liquid as one having a closed cup flash point below 100°F (37.8°C). A combustible liquid is defined, by DOT in the above reference, as one having a closed cup flash point at or above 100°F (37.8°C) and below 200°F (93.3°C). A pyrophoric liquid is defined, by DOT in the above reference, as one that ignites spontaneously in dry or moist air at or below 130°F (54.5°C). Materials with flash points of 200°F (93.3°C) or higher are not considered to be nonflammable or non-combustible, but are to be considered as burnable. The Environmental Protection Agency (EPA), in section 261.21 Subpart C, Part 261 of 40CFR*, 1982 Revised and as amended, designates the criteria for flammable and

*Refers only to materials that have become waste materials.

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combustible materials and oxidizers that exhibit the characteristic of ignitability (I). Liquids with closed cup flash points of less than 140°F (60°C) are defined by EPA as ignitable. The autoignition point (temperature) of a substance is generally defined as the minimum temperature required to initiate or cause self-sustained combustion in the absence of a spark or flame. Materials that ignite easily under normal industrial conditions are considered to be dangerous fire hazards. Such materials must be stored in a manner to prevent ignition and combustion. Easily ignitable substances, such as reducing agents, must be kept away from strong oxidizing agents. All containers must be tightly sealed. It is important to provide adequate ventilation in storage areas, and to locate the storage areas of these items away from fire hazards. Ample fire-control equipment must be easily accessible. Storage buildings, rooms and cabinets should comply with provisions of the National Fire Codes. The building should be electrically grounded and signs posted to prevent the lighting of matches or smoking in the area. Flammable storage areas should be equipped with smoke or fire detection equipment.

4.3.2.2 Water-sensitive fire and explosive hazardous materials. These are materials that react on contact with water or steam to ignite or evolve heat or explosive gases. Such materials exhibit the characteristic of reactivity (R) as designated by the EPA in Section 261.23 of the above reference.* These materials must be stored in well-ventilated, cool, dry areas. All containers must be tightly sealed. These materials are a fire hazard in contact with water or moisture; therefore, it is essential that no sprinkler be used. Otherwise, the building should conform to that required for storage of flammable materials. The building must be waterproof, located on high ground, and separated from other storage areas.

4.3.2.3 Incompatible materials. Materials that are chemically incompatible must be segregated in the storage of both serviceable and unserviceable items. The degree of segregation will depend upon DOD 4145.19-R-1 and local supplementary requirements that insure safe storage conditions. Hazardous storage compatibility codes are provided in the HMIS referred to in paragraph 4.2.

4.3.3 Chemical hazardous exposure limits. Chemical hazardous exposure limits for airborne concentrations of substances are obtained from the current TLVs® Threshold Limit Values for Chemical Substances in Workroom Air, adopted by the American Conference of Government Industrial Hygienists (ACGIH); current Occupational Safety and Health Administration (OSHA) Permissible Exposure Limits (PEL), 29 CFR, Section 1900.1000; and NIOSH Recommendation for Environmental Exposure Limits. Such information is also shown in MSDSs and the HMIS referred to in paragraph 4.2. Carcinogenic substances are listed by OSHA in Category I for confirmed carcinogens, and in Category II for suspected carcinogens. Category I substances have standard exposure limits set at the lowest possible levels. Category II substances have standard exposure limits set to prevent acute or chronic effects.

4.3.4 Toxicity. Toxicity information for chemical compounds is available from various publications and from MSDSs.

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4.3.4.1 EPA Toxic (T). Some chemical compounds have been designated by the EPA as toxic (T) in accordance with the criteria shown in Part 261, Subpart B, Section 261.11(a)(3) of 40CFR.* Some commercial chemical products are listed as toxic under Subpart D, Section 261.33(f).

4.3.4.2 EPA Acute Hazardous Toxicity (H). Some chemical compounds have been designated by the EPA as acute hazardous (H) in toxicity in accordance with the criteria shown in Subpart B, Section 261.11(a)(2) of the above reference.* Some commercial chemical products are listed as acute hazardous in toxicity under Subpart D, Section 261.33(e).

4.3.4.3 Hazardous toxic constituents. A list of chemical compounds and substances, shown to have toxic effects on humans or other life forms, is contained in Appendix VIII to 40CFR Part 261 and the Registry of Toxic Effects of Chemical Substances.

4.4 Pollution and disposal.

4.4.1 Pollution potential. All items described in this standard should be assumed to have a pollution potential. However, to minimize this potential, the proper use, storage and disposal methods must be strictly followed.

4.4.2 Disposal of excess or unserviceable material. To minimize disposal problems, it is recommended that no more than a one year's supply of each item listed in this standard be stocked. When stocks have been declared excess or unserviceable, they will be disposed of in accordance with the Defense Utilization and Defense Disposal Manual, DOD 4160.21-M, and applicable DOD Policy Memoranda and the Consolidated HM/HW Disposal Guidance. Guidance can be obtained from your servicing Defense Property Disposal Office (DPDO) on procedures required for proper reporting and turn-in.

4.4.3 Disposal and storage of hazardous wastes. Items are classified and managed as hazardous wastes as defined by the Resource Conservation and Recovery Act (RCRA) of 1976 (Public Law 94-580). Items have been identified as meeting the characteristics (i.e., ignitable, corrosive, reactive or EP toxic) or are listed (i.e., toxic or acute hazardous) according to Identification and Listing of Hazardous Waste, Part 261; 40CFR; or have been determined to be hazardous wastes by declaration of the Defense Property Disposal Service (DPDS) in accordance with procedures set forth in DOD 4160.21-M. Disposal of such items must be managed in accordance with the Installation Environmental Office, the DPDO, or the Safety and Health Office to insure proper reporting of disposal and treatment actions to the US EPA and State; and must be managed in accordance with Federal, State and local laws. The three main disposal methods are turn-in to the DPDO, on-post disposal by installation personnel, or disposal by commercial contract. Hazardous wastes that cannot be used, or disposed of as stated in paragraph 4.4.3.2, must be stored under environmentally safe conditions until suitable methods of disposal are determined. Short-term storage (less than 90 days) requires proper

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containment (i.e., packaging and facilities) in accordance with section 262.34, Part 262 of the above reference. Long-term storage (greater than 90 days) requires permitting by the EPA or by the state under Public Law 94-580 of 1976 (RCRA), in compliance with the requirements of 40CFR Parts 264 and 265. Physical custody will be accomplished by the activity with conforming storage or most nearly conforming storage. When physical custody is in question, the Post Commander will make the final decision. In all cases where the wastes are to be collected, stored, transported and disposed of at a state or local permitted disposal facility, the identity and description of the waste must be maintained and recorded in accordance with Part 262 of the above reference. Transportation of the waste must be in accordance with Part 263 of the above reference, Standards Applicable to Transporters of Hazardous Waste and title 49CFR, section 172.205, Hazardous Waste Manifesting.

4.4.3.1 Cleanup of liquid spills. To control the migration of spilled or leaking liquids, dike around the item with an inert, dry absorbent (e.g. clay, sawdust or vermiculite) or follow local spill plans. Control entry to the spill site and segregate salvageable materials away from the spill area. Initiate waste cleanup operations immediately in accordance with local procedures. The residue should be safety handled and transported to an approved or permitted disposal or storage facility. Packaging, labeling, transportation and record-keeping requirements for this waste material are determined by the appropriate Federal and State agencies and local procedures. It is recommended that all activities involving disposal preparation and transportation to commercial facilities be properly coordinated with the appropriate Federal and State agencies responsible for health and environmental aspects of hazardous materials. It is imperative that the proper description of waste accompany the packaged item at all times. Final disposal of the waste item should be accomplished by reutilization, transfer, donation or sales by DPDS in accordance with DOD 4160.21-M or by ultimate disposal as described in paragraph 4.4.3.2. Spill residue, including contaminants, to be turned in to the DPDO must first be properly identified, containerized, and labeled. For large scale spills that grossly contaminate the environment, the Chemical Transportation Emergency Center (CHEMTREC), can be called for assistance. Applicable procedures of the local spill control plan should be followed.

4.4.3.2 Ultimate disposal. Ultimate disposal shall be accomplished at a permitted or approved hazardous waste treatment or disposal facility designated by the Installation Environmental Office, DPDO, or Safety and Health Offices.

4.4.4 DISCLAIMER. RECOMMENDED DISPOSAL INSTRUCTIONS ARE FORMULATED FOR USE BY ELEMENTS OF THE DEPARTMENT OF DEFENSE. THE UNITED STATES OF AMERICA IN NO MANNER WHATSOEVER EITHER EXPLICITLY OR IMPLICITLY WARRANTS, STATES OR INTENDS SAID INSTRUCTION, TO HAVE ANY APPLICATION, USE OR VIABILITY BY OR TO ANY PERSON OR PERSONS CONTRACTING OUTSIDE THE DEPARTMENT OF DEFENSE OR ANY PERSON OR PERSONS CONTRACTING WITH ANY INSTRUMENTALITY OF THE UNITED STATES OF AMERICA AND DISCLAIMS ALL LIABILITY FOR SUCH USE. ANY PERSON USING THESE INSTRUCTIONS WHO IS NOT A MILITARY OR CIVILIAN EMPLOYEE OF THE UNITED STATES OF AMERICA SHOULD SEEK COMPETENT PROFESSIONAL ADVICE TO VERIFY AND ASSUME RESPONSIBILITY FOR THE SUITABILITY OF THESE

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INSTRUCTIONS TO THEIR PARTICULAR SITUATION REGARDLESS OF SIMILARITY TO A CORRESPONDING DEPARTMENT OF DEFENSE OR OTHER GOVERNMENT SITUATION.

5. DETAIL REQUIREMENTS

5.1 Ultraviolet and infrared spectrophotometry. Ultraviolet spectrophotometry is conducted in the electromagnetic spectrum region from 160 to 400 nanometers (nm) wavelength (WL), and infrared spectrophotometry is conducted in the region from 0.8 to 50 micrometers (μm). The transmittance of radiation is defined as the ratio of the radiant power transmitted through a sample to the radiant power transmitted through a reference material, where the detectors give an output proportional to the incident power. In practice, measurements are made instrumentally by comparing the radiant power transmitted by a solution containing an absorbing species to that of a "blank" solution, where the "blank" solution consists of the solvent and any other substances contained in the sample solution except the absorbing species. Since both solutions are measured in identical absorption cells, the radiant power transmitted by the "blank" solution corrects for absorption of radiation by the absorption cell and the substances other than the absorbing species. The transmittance "T" is expressed as: $T=P/P_0$; where "P" is the radiant power from the test solution sample and "P_0" is the radiant power from the "blank" solution reference sample reaching the detector. If "P_0" is set at 100 by instrumental adjustment for the wavelengths used for measurement, then the numerical value of "P", as measured directly, will be equal to the percent transmittance for the test solution sample. The absorbance "A" is defined as the logarithm of the reciprocal of transmittance, and is expressed as: $A = \log 1/T = \log P_0/P$, and $A = abc = \varepsilon bc/(MW)$; where "b" is the radiation path length in cm, "c" is the concentration in grams per liter, "MW" is the molecular weight, "a" is the absorptivity, and "ε" is the molar absorptivity. Theoretically, absorbances can vary from zero to infinity, but absorbances greater than two or three are seldom useable in practice. Absorption spectra can be shown graphically as absorbance or transmittance plotted against increasing values of wavelength or decreasing values of frequency in the form of wave numbers (cm^{-1}). Ultraviolet spectra are usually determined on samples in the form of dilute liquid solutions. Solvents for ultraviolet spectrophotometry can be selected on the basis of low absorbance (high transmittance) in the desired wavelength region. Also since absorbances are additive, subtraction of the contribution due to the solvent from the observed absorbance of a solution enables the absorbance curve of a solute substance to be determined. Infrared spectra of liquid samples are usually determined for the neat forms, but solid solutes are dissolved and examined as a dilute solution in a solvent which is nonabsorbing in the regions of interest. Infrared spectra of powders or solid particles can be obtained for slurries or mulls in a liquid medium of approximately the same refractive index. Solvents for infrared spectrophotometry can be selected on the basis of the wavelength bands desired for determination of the spectra of a solute substance.

5.2 Spectrophotometric grade solvents. Spectrophotometric grade solvents are at least as pure as analytical reagent grade chemicals, and in addition, meet specified requirements for ultraviolet absorbance and

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infrared transmittance. Specifications for such solvents include absorbance data in the ultraviolet region, and also bands of maximum infrared transmittance (infrared windows) at 50 to 100% T. American Chemical Society (ACS) reagent chemical solvents, covered by Federal specification O-C-265, that are suitable for use in ultraviolet spectrophotometry are specified by the absorbance of the sample measured in a 1.00 cm cell against water, as the reference liquid, in a similar matched cell set at zero absorbance. The absorbance, ranging from 1.00 maximum (at cutoff) to 0.01, is specified in terms of specific wavelengths for each solvent compound; and the spectral absorbance curve recorded through the indicated wavelengths should be smooth throughout the prescribed range and should not show any extraneous impurity peaks within this range. Spectrographic grade solvents, other than ACS reagent grade, are specified by their manufacturers in a similar manner for ultraviolet absorbance. There are no ACS specifications for infrared transmittance bands of spectrophotometric grade solvents, but manufacturers of these compounds supply such data for most of them. A list of available spectrophotometric solvents from several sources is shown in Table I. This list shows the absorbance (A max) for specific wavelengths in the ultraviolet region for each solvent with the cutoff point at A=1.00 which corresponds to a transmittance of 10%, and ranging down to an absorbance of A=0.01 which corresponds to a transmittance of 97.75%. The spectral absorbance curves for the wavelengths indicated shall be smooth throughout the prescribed range and shall not show any extraneous impurity peaks within this range. This list also shows the infrared wavelength bands for 50 to 100% transmittance (T) of these solvents where such data was available from the manufacturers.

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TABLE I. SPECTROPHOTOMETRIC GRADE SOLVENTS

Compound	Assay (% min)	Ultraviolet Absorbance A (max)	Ultraviolet Absorbance WL (nm)	Infrared Wavelength Bands (50 to 100% T) Path (mm)	Infrared Wavelength Bands (50 to 100% T) WL (μm)
Acetic Acid, Glacial CH_3COOH	99.7	1.00 0.15 0.02 0.01	255 260 280 300-400	---	---
Acetone, ACS $(CH_3)_2CO$	99.5	1.00 0.10 0.02 0.01	330 340 350 400	0.1	2.0-3.2 3.6-5.5 8.6-8.9 9.4-10.7 11.4-12.7 12.8-15.0
Acetonitrile CH_3CN	98.0	1.00 0.15 0.075 0.01 0.005	210 220 230 240 250-400	0.1	2.0-3.1 3.5-4.3 4.5-6.5 7.6-9.3 9.9-10.8 11.0-13.2 13.6-16.2
Benzene, ACS C_6H_6	99.0	1.00 0.30 0.10 0.04 0.02 0.01	280 290 300 330 350 380-400	0.1	2.0-3.2 3.4-5.0 5.2-5.4 5.6-6.5 6.9-7.1 7.3-8.4 8.6-9.5 10.0-13.8
Benzonitrile C_6H_5CN	98.0	1.00 0.20 0.09 0.05 0.03 0.02 0.01	300 310 320 330 340 350 360	0.03	2.0-3.2 3.3-4.4 4.5-6.2 6.3-6.7 7.0-7.7 7.8-8.5 8.6-9.3 9.4-9.6 9.8-10.8 10.9-13.0 13.5-14.2 15.0-16.0

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TABLE I. SPECTROPHOTOMETRIC GRADE SOLVENTS - Continued.

Compound	Assay (% min)	Ultraviolet Absorbance A (max)	Ultraviolet Absorbance WL (nm)	Infrared Wavelength Bands (50 to 100% T) Path (mm)	Infrared Wavelength Bands (50 to 100% T) WL (μm)
Benzonitrile (Continued)				0.1	2.0-3.2 3.4-4.4 4.6-5.0 5.3-6.1 7.8-8.3 8.7-9.0 10.0-10.5 11.0-11.7 11.9-12.4
Bromoform $CHBr_3$	----	1.00 0.50 0.20 0.02 0.01 <0.01	335 340 350 375 400 400-450	0.1	2.5-3.2 3.3-8.5 9.0-14.1
Bromotrichloro-methane $BrCCl_3$	97.0	1.00 0.45 0.15 0.05 0.02 0.01 <0.01	340 350 360 370 380 390 400-450	0.1	2.0-6.3 6.9-10.2 10.6-12.1
Butyl Acetate $CH_3COO(CH_2)_3CH_3$	98.0	1.00 0.30 0.10 0.05 0.03 0.02 0.01 <0.01	255 260 270 280 300 310 320 330-400	0.03 0.1	2.0-3.2 3.6-5.5 6.1-6.6 12.6-13.3 13.8-14.6 2.0-3.2 3.6-5.3 12.5-13.0 13.6-15.0
Butyl Alcohol (1-Butanol) $CH_3(CH_2)_2CH_2OH$	98.0	1.00 0.30 0.15 0.06 0.03 0.01 <0.01	210 220 230 240 250 260 270-400	0.03 0.1	2.0-2.8 3.2-6.7 8.3-8.8 10.7-11.7 12.0-13.0 2.0-2.8 4.3-6.0

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TABLE I. SPECTROPHOTOMETRIC GRADE SOLVENTS - Continued.

Compound	Assay (% min)	Ultraviolet Absorbance A (max)	Ultraviolet Absorbance WL (nm)	Infrared Wavelength Bands (50 to 100% T) Path (mm)	Infrared Wavelength Bands (50 to 100% T) WL (μm)
Butyl Ether (Di-n-butylether) $[CH_3(CH_2)_3]_2O$	99.0	1.00	220	---	---
		0.75	225		
		0.50	245		
		0.10	265		
		0.01	310		
Carbon Disulfide CS_2	99.0	1.00	380	0.1	2.0-4.3
		0.90	383		4.8-6.2
		0.50	388		7.1-16.0
		0.25	394		
		0.10	403		
		0.05	410		
		0.01	500-750		
Carbon Tetra-chloride, ACS CCl_4	99.0	1.00	265	0.1	2.0-12.1
		0.35	270		12.7-13.0
		0.10	280		14.0-15.0
		0.05	290		
		0.02	300		
		0.01	330-400		
Chlorobenzene C_6H_5Cl	99.0	1.00	287	---	---
		0.30	290		
		0.07	300		
		0.01	360-400		
Chloroform, ACS $CHCl_3$	98.0	1.00	245	0.1	2.0-3.3
		0.25	255		3.4-8.0
		0.15	260		8.4-12.2
		0.05	270		
		0.01	290-400		
Cyclohexane, ACS $CH_2(CH_2)_4CH_2$	98.5	1.00	210	0.1	2.0-3.3
		0.50	220		3.9-6.7
		0.20	230		7.1-7.9
		0.08	240		8.1-9.5
		0.03	250		10.0-11.0
		0.02	260		11.8-15.0
		0.01	300-400		
Cyclopentane $CH_2CH_2CH_2CH_2CH_2$	98.0	1.00	198	---	---
		0.50	210		
		0.10	220		
		0.01	240-400		

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TABLE I. SPECTROPHOTOMETRIC GRADE SOLVENTS

Compound	Assay (% min)	Ultraviolet Absorbance A (max)	Ultraviolet Absorbance WL (nm)	Infrared Wavelength Bands (50 to 100% T) Path (mm)	Infrared Wavelength Bands (50 to 100% T) WL (μm)
Decahydronaphthalene $C_{10}H_{18}$	98.0	1.00	200	0.03	2.0-3.3
		0.85	210		3.8-6.7
		0.60	220		7.1-7.7
		0.32	230		7.7-8.8
		0.15	240		9.0-10.2
		0.05	250		10.4-10.8
		0.01	260		10.9-11.6
		<0.01	270-400		12.3-16.0
o-Dichlorobenzene $C_6H_4Cl_2$	99.0	1.00	295	---	---
		0.30	300		
		0.10	305		
		0.05	335		
		0.01	375-400		
1,2-Dichloroethane, ACS (Ethylene dichloride) CH_2ClCH_2Cl	99.0	1.00	226	0.1	2.0-3.3
		0.50	230		3.4-6.7
		0.20	235		7.2-7.4
		0.10	240		8.3-9.6
		0.05	245		9.8-10.4
		0.02	250		11.6-12.7
		0.01	255-400		
Dichloromethane, ACS (Methylene chloride) CH_2Cl_2	99.0	1.00	235	0.1	2.0-3.2
		0.35	240		3.4-4.3
		0.10	250		4.4-6.8
		0.04	260		7.1-7.7
		0.01	340-400		8.1-11.1
					11.3-12.8
Diethyl Ketone (3-Pentanone) $CH_3CH_2COCH_2CH_3$	98.0	1.00	330	---	---
		0.15	340		
		0.05	350		
		0.01	370-400		
1,2-Dimethoxyethane (Ethylene glycol dimethyl ether) $CH_3OCH_2CH_2OCH_3$	99.0	1.00	240	---	---
		0.35	250		
		0.20	260		
		0.15	270		
		0.05	280		
		0.01	300		
		0.00	330		
N,N-Dimethylacetamide $CH_3CON(CH_3)_2$	98.0	1.00	268	0.03	2.0-3.2
		0.30	280		3.6-5.7
		0.18	290		10.6-13.5
		0.12	300		13.7-15.0

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TABLE I. SPECTROPHOTOMETRIC GRADE SOLVENTS - Continued.

Compound	Assay (% min)	Ultraviolet Absorbance A (max)	Ultraviolet Absorbance WL (nm)	Infrared Wavelength Bands (50 to 100% T) Path (mm)	Infrared Wavelength Bands (50 to 100% T) WL (μm)
N,N-Dimethyl-acetamide (Continued)		0.08	310	0.1	2.0-3.2
		0.05	320		3.9-5.0
		0.03	330		5.2-6.5
		0.02	340		12.6-16.0
		<0.005	350-400		
N,N-Dimethyl-formamide, ACS (Dimethylformamide) $HCON(CH_3)_2$	99.0	1.00	270	0.1	2.0-2.7
		0.30	275		3.0-3.3
		0.10	295		3.6-5.7
		0.05	310		8.1-8.6
		0.01	340-400		9.6-11.5
					11.6-14.5
p-Dioxane (1,4-Dioxane) $CH_2CH_2OCH_2CH_2O$	99.0	1.00	215	0.1	2.0-3.3
		0.70	216		3.8-5.0
		0.50	225		5.1-6.7
		0.24	250		9.9-10.7
		0.10	270		12.1-15.4
		0.05	280		
		0.01	295-400		
Dodecane $CH_3(CH_2)_{10}CH_3$	98.0	1.00	200	0.1	2.0-3.3
		0.40	210		4.1-6.7
		0.15	220		7.5-13.6
		0.07	230		14.1-16.0
		0.04	240		
		0.03	250		
		0.02	260		
		0.01	280		
		<0.01	290-400		
Ethyl Acetate, ACS $CH_3COOCH_2CH_3$	99.0	1.00	255		2.0-3.2
		0.50	257		3.6-5.3
		0.10	263		11.1-11.6
		0.05	275		13.0-15.0
		0.01	330-400		
Ethyl Alcohol, ACS (Ethanol) CH_3CH_2OH	95.0	0.40	210	---	---
		0.25	220		
		0.15	230		
		0.05	240		
		0.01	270-400		

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TABLE I. SPECTROPHOTOMETRIC GRADE SOLVENTS - Continued.

Compound	Assay (% min)	Ultraviolet Absorbance A (max)	Ultraviolet Absorbance WL (nm)	Infrared Wavelength Bands (50 to 100% T) Path (mm)	Infrared Wavelength Bands (50 to 100% T) WL (μm)
Ethyl Ether (Diethyl Ether) $(CH_3CH_2)_2O$	99.0*	1.00	220	0.03	2.6-3.3
		0.40	230		4.1-6.7
		0.10	250		7.5-8.4
		0.03	260		9.9-11.8
		<0.01	280-400		12.0-16.0
*Exclusive of preservative.					
Formamide $HCONH_2$	99.0	1.00	265	0.025	3.9-5.5
		0.20	270		10.3-10.8
		0.08	280		
		0.04	290		
		0.02	300		
		<0.01	310-400		
Glycerol (Glycerine) $CH_2OHCHOHCH_2OH$	99.5	1.00	205	0.03	2.0-2.8
		0.70	210		4.4-6.0
		0.35	220	0.1	2.0-2.7
		0.18	230		3.9-9.5
		0.07	250		
		0.05	270		
		0.02	300		
		<0.01	350-400		
Heptane $CH_3(CH_2)_5CH_3$	98.5	1.00	210	0.1	2.0-3.3
		0.20	220		3.8-6.6
		0.05	230		7.5-12.8
		0.02	240		13.1-13.3
		0.01	250		14.1-16.0
		<0.01	260-400		
Hexadecane $CH_3(CH_2)_{14}CH_3$	99.0	1.00	210	---	---
		0.50	220		
		0.25	235		
		0.05	260		
		0.01	310		
Hexanes, ACS (n-Hexane or a mixture of isomers of hexane (C_6H_{14}) predominantly n-hexane containing methylcyclopentane (C_6H_{12}))		1.00	210	0.1	2.0-3.3
		0.20	220		3.6-6.6
		0.10	230		7.3-13.6
		0.04	240		14.1-16.0
		0.02	250		
		0.01	280-400		

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TABLE I. SPECTROPHOTOMETRIC GRADE SOLVENTS - Continued.

Compound	Assay (% min)	Ultraviolet Absorbance A (max)	Ultraviolet Absorbance WL (nm)	Infrared Wavelength Bands (50 to 100% T) Path (mm)	Infrared Wavelength Bands (50 to 100% T) WL (μm)
Isobutyl Alcohol, ACS (2-Methyl-1-propanol) $(CH_3)_2CH_2CH_2OH$	99.0	1.00 0.25 0.07 0.05 0.01	230 240 250 260 330-400	0.1	2.0-2.7 4.1-6.4
Isopropyl Alcohol, ACS (2-Propanol) $(CH_3)_2CHOH$	99.0	1.00 0.40 0.20 0.08 0.04 0.03 0.02 0.01	210 220 230 245 260 275 300 330-400	0.1	2.0-2.7 3.6-6.8 8.0-9.0 9.3-10.4 10.8-12.2 12.5-13.4
2,6-Lutidine (2,6-Dimethylpyridine) $N{:}C(CH_3)CH{:}CHCH{:}C(CH_3)$	98.0	1.00 0.50 0.15 0.07 0.03 <0.01	303 305 310 320 330 360-400	0.03	2.0-3.2 3.6-6.2 7.5-7.8 8.3-8.6 8.8-9.1 10.4-12.5 13.4-13.7 14.1-15.0
Methanol, ACS (Methyl Alcohol) CH_3OH	99.5	1.00 0.80 0.40 0.20 0.10 0.04 0.01	205 210 220 230 240 260 280-400	Film	2.0-2.8 3.5-9.5 10.0-14.5
2-Methoxyethanol (Ethylene Glycol Monomethyl Ether) $CH_3OCH_2CH_2OH$	98.0	1.00 0.55 0.40 0.27 0.15 0.07 0.05 0.03 0.02 0.01 <0.01	215 220 230 240 250 260 270 280 290 300-330 330-400	0.03 0.1	2.0-2.9 3.9-6.5 2.5-2.8 4.1-5.1 5.2-6.4

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TABLE I. SPECTROPHOTOMETRIC GRADE SOLVENTS - Continued.

Compound	Assay (% min)	Ultraviolet Absorbance A (max)	Ultraviolet Absorbance WL (nm)	Infrared Wavelength Bands (50 to 100% T) Path (mm)	Infrared Wavelength Bands (50 to 100% T) WL (μm)
2-Methylbutane (Isopentane) $(CH_3)_2CHCH_2CH_3$	99.0	1.00 0.15 0.10 0.05 0.01	192 210 223 233 250	---	---
Methylcyclohexane $CH_3CH(CH_2)_4CH_2$	99.0	1.00 0.36 0.14 0.06 0.03 0.01 <0.01	210 220 230 240 250 260 270-400	0.1	2.0-3.3 3.6-6.6 7.5-7.9 8.0-10.2 10.5-11.0 11.1-11.4 11.5-11.8 12.0-16.0
Methyl Ethyl Ketone (2-Butanone) $CH_3COCH_2CH_3$	99.0	1.00 0.40 0.10 0.02 0.01	330 335 340 345 350-400	0.1	2.0-2.8 2.9-3.2 3.6-5.5 9.4-9.8 11.0-12.8 13.5-15.0
Methyl Formate $HCOOCH_3$	98.0	1.00 0.05 0.04 0.03 0.02 0.01 <0.01	260 270 280-290 300 310 320 330-400	0.03	2.0-3.3 3.5-5.6 6.1-6.8 7.3-7.9 9.2-10.8 11.3-13.0 13.2-16.0
Methyl Isobutyl Ketone (4-Methyl-2-Pentanone) $CH_3COCH_2CH(CH_3)_2$	99.0	1.00 0.50 0.25 0.15 0.02 0.01	335 340 350 360 380 400	0.1	2.0-3.2 3.6-5.5 6.2-6.4 9.2-9.9 10.9-11.7 12.0-15.0
Methylpyrrolidone (1-Methyl-2-Pyrrolidinone) $CH_3N(CH_2)_3CO$	98.0	1.00 0.55 0.25 0.12 0.06 0.03 0.02 0.01 <0.01	265 270 280 290 300 310 320 330 340-400	0.03 0.1	2.0-3.3 3.6-5.8 6.2-6.6 7.2-7.6 8.3-8.8 9.2-10.1 10.3-15.0 2.0-2.3 2.4-3.0 3.7-5.5 12.3-13.0

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TABLE I. SPECTROPHOTOMETRIC GRADE SOLVENTS - Continued.

Compound	Assay (% min)	Ultraviolet Absorbance A (max)	Ultraviolet Absorbance WL (nm)	Infrared Wavelength Bands (50 to 100% T) Path (mm)	Infrared Wavelength Bands (50 to 100% T) WL (μm)
Methyl Sulfoxide (Dimethyl Sulfoxide) $(CH_3)_2SO$	98.0	1.00	265	0.03	2.0-3.3
		0.40	270		3.6-6.9
		0.25	280		7.8-9.0
		0.18	290		11.3-14.0
		0.10	300	0.1	2.0-2.3
		0.06	310		2.4-3.2
		0.03	320		3.7-5.0
		0.02	330		5.1-6.7
		0.01	340		7.9-8.7
		<0.01	350-400		11.4-13.8
Nitromethane CH_3NO_2	95.0	1.00	380	0.1	2.0-3.2
		0.25	390		3.5-4.0
		0.05	400		4.1-6.1
		0.01	410		9.6-10.9
		<0.01	420-450		11.1-14.3
Pentane $CH_3(CH_2)_3CH_3$	99.0	1.00	205	0.1	2.0-3.3
		0.30	210		3.7-6.6
		0.10	220		7.5-10.8
		0.01	245-400		11.1-11.5
					11.7-13.5
					14.0-16.0
Pyridine N:CHCH:CHCH:CH	98.0	1.00	305	0.1	2.0-3.2
		0.50	310		3.4-6.1
		0.25	320		6.5-6.7
		0.15	330		7.1-8.1
		0.01	340		8.3-8.6
		<0.01	360-400		8.9-9.3
					10.3-12.8
Tetrachloroethylene $Cl_2C=CCl_2$	99.0	1.00	300	0.1	2.0-8.7
		0.25	305		8.8-10.4
		0.15	310		11.4-12.0
		0.05	350		13.4-16.0
		0.03	400		
Tetrahydrofuran $OCH_2CH_2CH_2CH_2$	99.0	1.00	225	---	---
		0.50	240		
		0.25	250		
		0.08	270		
		0.02	290		
		0.01	230-400		

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TABLE I. SPECTROPHOTOMETRIC GRADE SOLVENTS - Continued.

Compound	Assay (% min)	Ultraviolet Absorbance A (max)	Ultraviolet Absorbance WL (nm)	Infrared Wavelength Bands (50 to 100% T) Path (mm)	Infrared Wavelength Bands (50 to 100% T) WL (μm)
Toluene, ACS $C_6H_5CH_3$	99.0	1.00 0.50 0.20 0.10 0.05 0.02 0.01	286 288 293 300 310 335 350-400	0.1	2.0-3.2 3.6-5.1 5.2-5.4 5.6-6.1 7.4-8.4 8.6-9.0 9.9-11.1 11.3-12.7 12.9-13.2
1,1,2-Trichloro-1,2,2-Trifluoroethane Cl_2FCCF_2Cl	99.0	1.00 0.20 0.01	231 240 260-400	0.1	2.0-7.9 13.0-14.9
2,2,2-Trifluoroethanol F_3CCH_2OH	99.0	1.00 0.50 0.20 0.12 0.03 0.01	190 195 200 210 235 260	---	---
2,2,4-Trimethylpentane, ACS (Isooctane) $(CH_3)_3CCH_2CH(CH_3)_2$	99.0	1.00 0.20 0.10 0.04 0.01	210 220 230 240 250-400	0.1	2.0-3.3 3.6-6.7 7.5-8.0 8.6-10.2 10.3-16.0
m-Xylene $C_6H_4(CH_3)_2$	----	1.00 0.40 0.20 0.10 0.07 0.05 0.03 0.02 0.01	290 300 310 320 330 340 350 360 370-400	0.03	2.0-3.2 3.6-6.2 7.4-8.5 8.7-9.1 9.3-9.5 9.9-11.0 11.6-12.7 13.5-14.2
o-Xylene $C_6H_4(CH_3)_2$	98.0	1.00 0.15 0.07 0.03 0.01	300 310 320 340 360-400	0.1	2.0-3.1 3.6-5.2 5.6-5.9 7.4-8.3 8.6-8.9 10.3-10.8 11.5-12.3 15.2-16.0

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Assignee: GS

Custodians:
Army - EA
Navy - SH
Air Force - 68

Review Activities:
Army - MD

Preparing Activity:
Army - EA

Civil Agency Coordinating Interest:
GSA
NASA - MSF

Project Number 6810-B397

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